Question Slides

LaTeXQuest

2026-10-15

Question 1

Example 1. Find the value of

(i) $\frac{25!}{23!}$ (ii) $\frac{9!}{6! \times 3!}$ (iii) $(7-4)!$

(iv) $7! - 5!$ (v) $2! + 3!$.

[1.0 mark]

Solution 1

No solution provided.

Question 2

Example 2. Evaluate $\frac{n!}{(n-r)!}$ when
(i) $n = 9$, $r = 5$ (ii) $r = 3$.

[1.0 mark]

Solution 2

No solution provided.

Question 3

Example 3. Evaluate $\dfrac{n!}{r!\,(n-r)!}$ when

(i) $n = 7,\ r = 5$ (ii) $r = 2.$

[1.0 mark]

Solution 3

No solution provided.

Question 4 – Algebra

Example 4. Convert the following products to factorials:
(i) $5 \times 6 \times 7 \times 8 \times 9$
(ii) $2 \times 4 \times 6 \times 8 \times 10 \times 12$.

[1.0 mark]

Solution 4

No solution provided.

Question 5

Example 5. Find n, if

(i) $\frac{1}{8!} + \frac{1}{9!} = \frac{n}{10!}$

(ii) $(n+1)! = 12 \times (n-1)!$

[1.0 mark]

Solution 5

No solution provided.

Question 6

Example 6. Find n:

(i) if $\dfrac{n!}{2!\,(n-2)!}$ and $\dfrac{n!}{4!\,(n-4)!}$ are in the ratio $2:1$

(ii) if $\dfrac{n!}{2!\,(n-2)!}$ and $\dfrac{(2n)!}{3!\,(2n-3)!}$ are in the ratio $3:44$.

[1.0 mark]

Solution 6

No solution provided.

Question 7

Example 7. Prove that $(2n)! = 1 \cdot 3 \cdot 5 \cdot \ldots \cdot (2n-1) \cdot 2^n \cdot n!$.

[1.0 mark]

Solution 7

No solution provided.

Question 8

Example 8. Find H.C.F. and L.C.M. of 5!, 6! and 7!.

[1.0 mark]

Solution 8

No solution provided.